

Demonstration of Proposed Features

Date	02 July 2026
Team ID	SWTID-2026-3218
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the developer's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Impl/Partial/Pending)	Demonstrated (Yes/No)	Remarks
1	Data Collection & Preprocessing	Load credit card dataset from public source, handle missing values, encode categorical features (gender, industry, citizenship).	Implemented	Yes	Successfully loaded UCI ML repository dataset and performed feature engineering.
2	XGBoost Model Training	Train XGBoost classifier on preprocessed data with hyperparameter tuning using cross-validation.	Implemented	Yes	Model achieved 90% accuracy, 0.9453 AUC, saved as joblib artifact.
3	Flask REST API Backend	Build /predict endpoint to accept applicant data and return approval decision with confidence score.	Implemented	Yes	Endpoint tested with Postman; returns JSON with approved status and confidence %.
4	Web UI with Form Inputs	Create responsive HTML form with 12 input fields (age, income, debt, credit score, employment details, citizenship, industry).	Implemented	Yes	Form validates inputs, displays success/error messages with approx icons.
5	Real-Time Predictions	Submit applicant details via form and receive instant approval/rejection with confidence percentage.	Implemented	Yes	Predictions return within milliseconds with 85-95% confidence scores.

6	Categorical Encoding with Fallback	Implement safe_encode() function to handle unseen category values gracefully without breaking the API.	Implemented	Yes	Fallback logic tested; unknown values default to index 0, prevents runtime errors.
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Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%